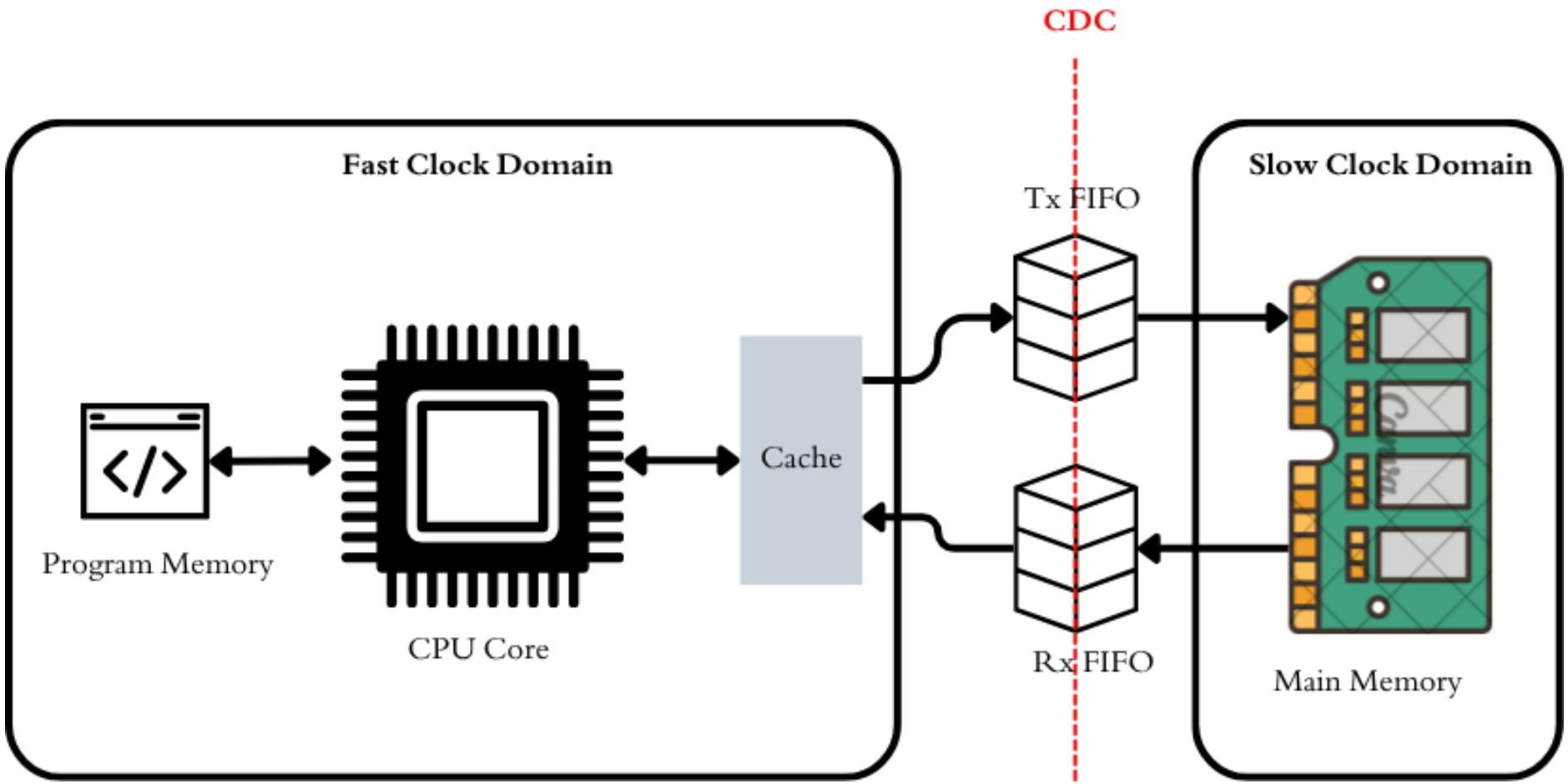
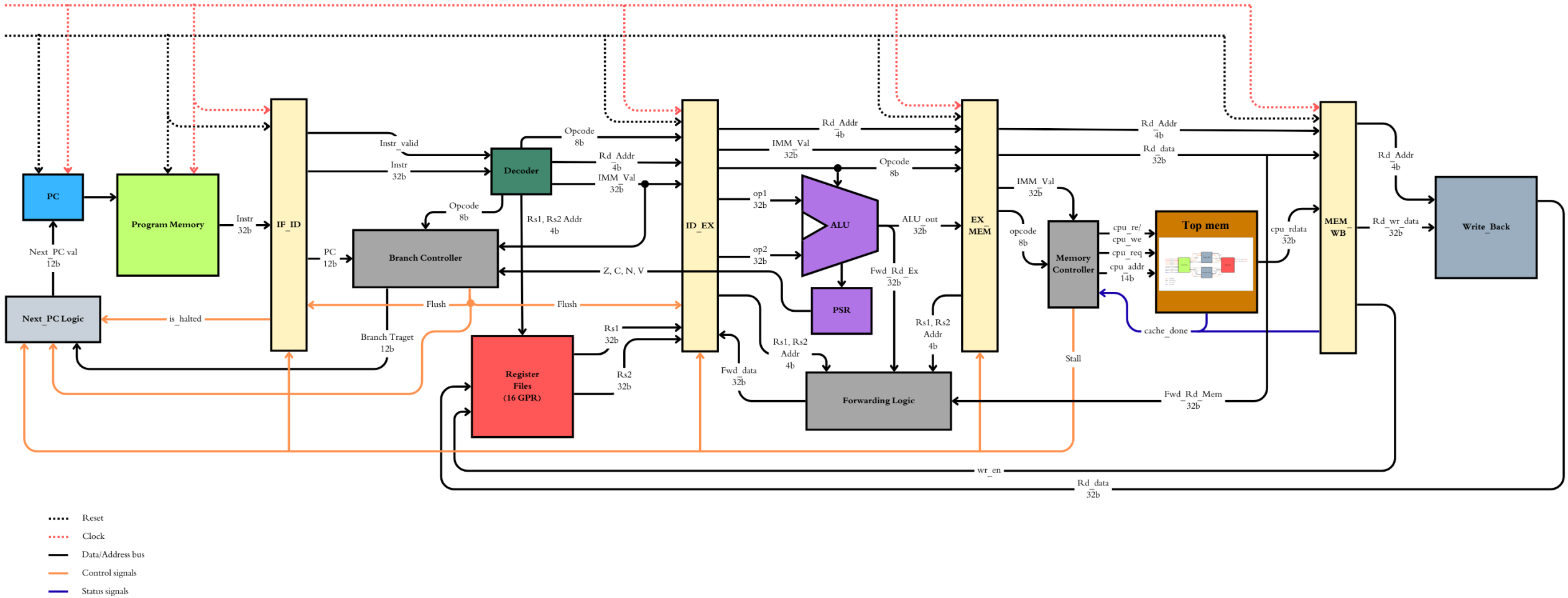
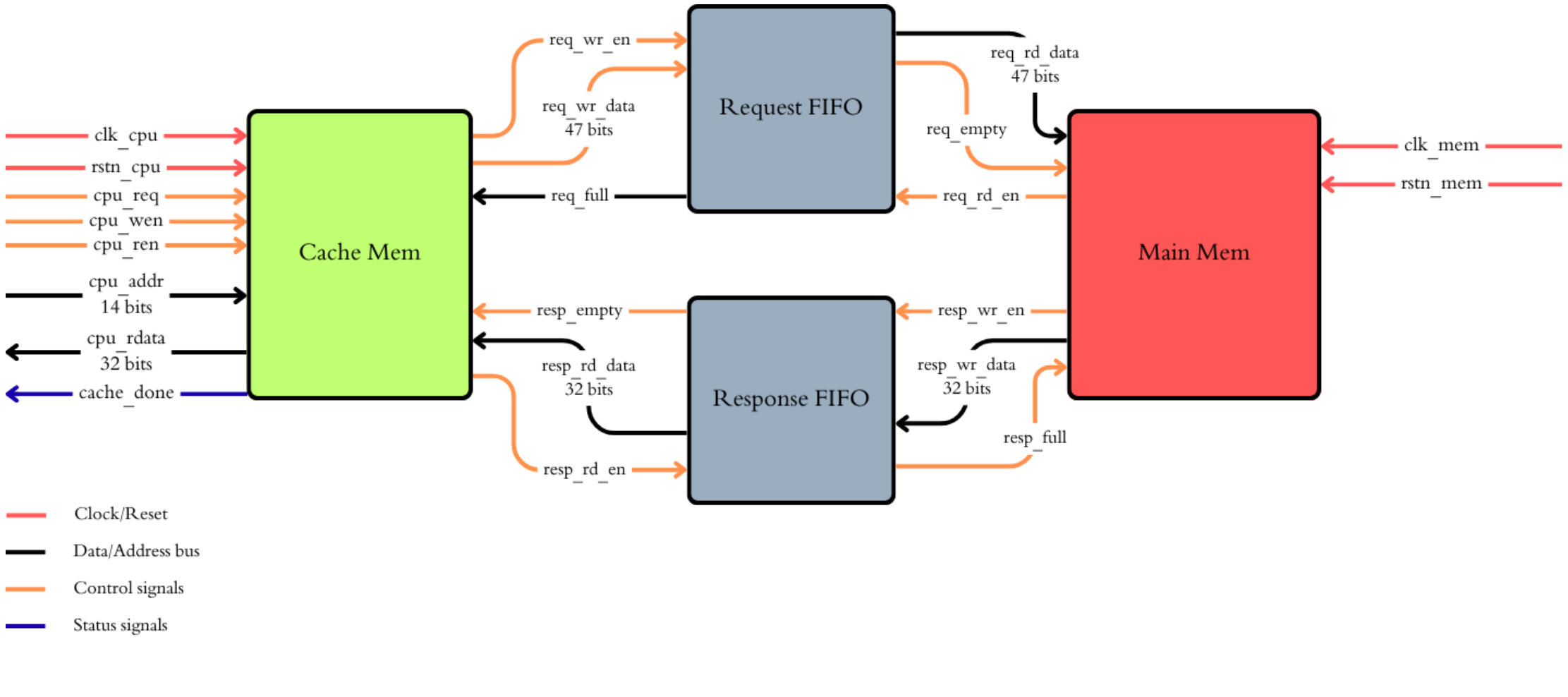
CPU_Lite


CDC
Fast Clock Domain
Slow Clock Domain
Tx FIFO
Cache
Program Memory
CPU Core
Rx FIFO
Main Memory


PC
Program Memory
Instr
32b
IF_ID
Instr_valid
Instr
32b
Decoder
Opcode
8b
Rd_Addr
4b
IMM_Val
32b
Opcode
8b
Rs1, Rs2 Addr
4b
Next_PC val
12b
Next_PC Logic
is_halted
PC
12b
Branch Controller
Flush
Flush
Branch Traget
12b
Z, C, N, V
Register
Files
(16 GPR)
Rs1
32b
Rs2
32b
ID_EX
op1
32b
op2
32b
ALU
ALU_out
32b
PSR
Fwd_Rd_Ex
32b
Rs1, Rs2
Addr
4b
Fwd_data
32b
Forwarding Logic
EX_
MEM
Rs1, Rs2
Addr
4b
Rd_Addr
4b
Rd_data
32b
IMM_Val
32b
opcode
8b
Memory
Controller
Stall
cpu_re/
cpu_we
cpu_req
cpu_addr
14b
Top mem
cache_done
cpu_rdata
32b
Fwd_Rd_Mem
32b
MEM_
WB
Rd_Addr
4b
Rd_wr_data
32b
Write_Back
wr_en
Rd_data
32b
Reset
Clock
Data/Address bus
Control signals
Status signals


clk_cpu
rstn_cpu
cpu_req
cpu_wen
cpu_ren
cpu_addr
14 bits
cpu_rdata
32 bits
cache_done
Cache Mem
req_wr_en
req_wr_data
47 bits
req_full
Request FIFO
req_rd_data
47 bits
req_empty
req_rd_en
Main Mem
clk_mem
rstn_mem
resp_empty
resp_rd_data
32 bits
resp_rd_en
Response FIFO
resp_wr_en
resp_wr_data
32 bits
resp_full
Clock/Reset
Data/Address bus
Control signals
Status signals